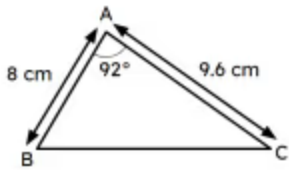


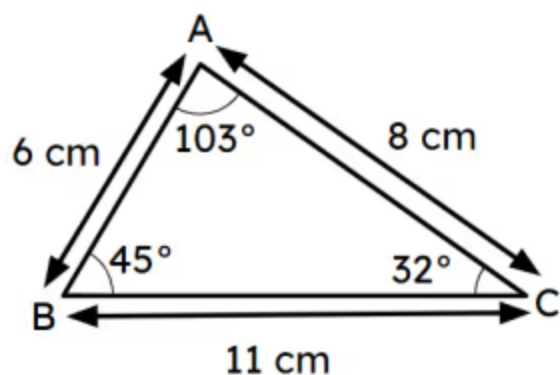
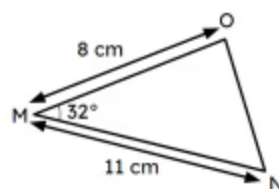
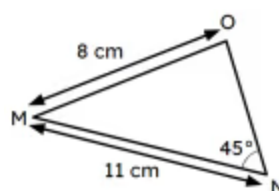
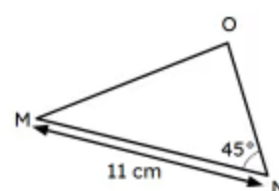
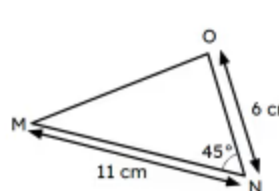


Congruent triangles (SAS)

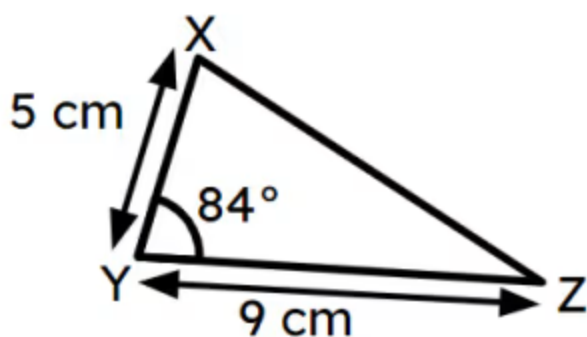
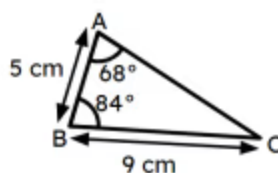
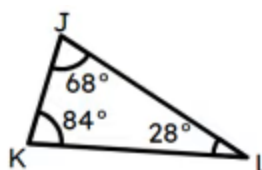
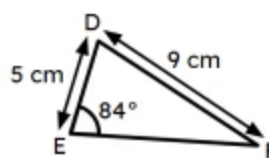
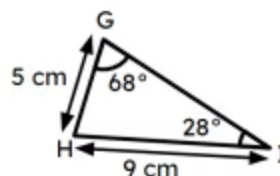
- 1 When there are two triangles, if the two corresponding edges and the angle between them are the same, you can prove they are congruent by _____ (side-angle-side). Fill in the blank
- 2 If triangles ABC and XYZ are congruent by SAS then _____ is 9.6 cm. Fill in the blank



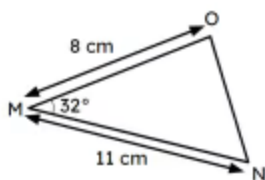
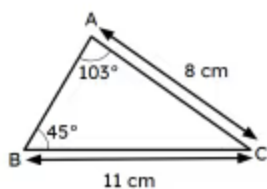
3 Which of the following triangles have sufficient information to guarantee they are congruent to triangle ABC? Tick 2 correct answers


☐

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☐


4 Which of the following triangles are congruent to triangle XYZ? Tick 2 correct answers


☐

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☐

☐


5 Are the triangles ABC and MNO congruent? Tick 1 correct answer


☐

Yes, they are guaranteed to be congruent.

☐

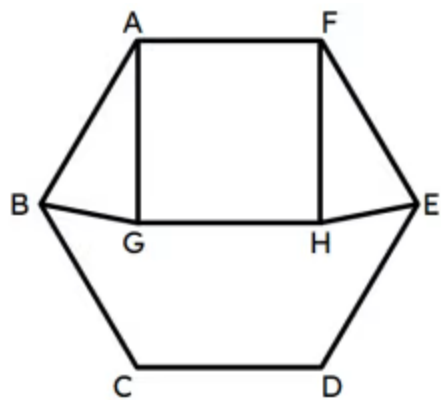
No, they are definitely incongruent.

☐

There is not enough information to tell.



6 This diagram shows a regular hexagon ABCDEF and a square AGHF. Match the statements to prove triangle ABG and FHE are congruent by SAS. Write the correct letter in each box



a	$\angle BAF = \angle EFA$
b	$\angle GAF = \angle HFA$
c	$\angle BAG =$
d	$\angle HFE =$
e	$\angle BAG =$
f	$AB = AG = FH = FE$

	as they are interior angles of a regular hexagon ($= 120^\circ$)
	$\angle EFA - \angle HFA (= 30^\circ)$
	as they are in regular polygons with shared edges
	as they are interior angles of a square ($= 90^\circ$)
	$\angle BAF - \angle GAF (= 30^\circ)$
	$\angle HFE$